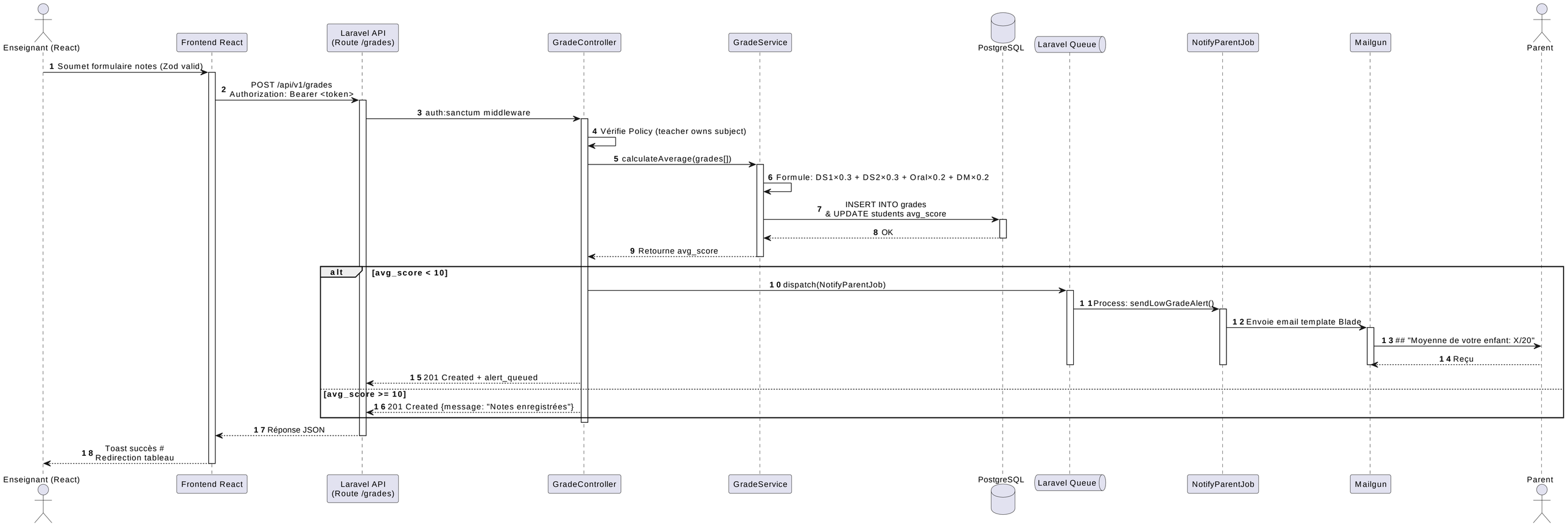
Enseignant (React)
Frontend React
Laravel API (Route /grades)
GradeController
GradeService
PostgreSQL
Laravel Queue
NotifyParentJob
Mailgun
Parent
1 Soumet formulaire notes (Zod valid)
2 POST /api/v1/grades Authorization: Bearer <token>
3 auth:sanctum middleware
4 Vérifie Policy (teacher owns subject)
5 calculateAverage(grades[])
6 Formule: DS1×0.3 + DS2×0.3 + Oral×0.2 + DM×0.2
7 INSERT INTO grades & UPDATE students avg_score
8 OK
9 Retourne avg_score
alt
[avg_score < 10]
10 dispatch(NotifyParentJob)
11 Process: sendLowGradeAlert()
12 Envoie email template Blade
13 ## "Moyenne de votre enfant: X/20"
14 Reçu
15 201 Created + alert_queued
[avg_score >= 10]
16 201 Created {message: "Notes enregistrées"}
17 Réponse JSON
18 Toast succès # Redirection tableau